

The Sign-Constrained Spherical Perceptron: Null Control for the Tempotron Capacity Program

Tempotron learning-surface program

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Abstract

We derive the storage capacity of the *sign-constrained* (excitatory-only, $\omega_i \geq 0$, Dale’s law) spherical perceptron using the Gardner replica-symmetric (RS) free energy. The positive-orthant constraint modifies only the entropic coefficient of the RS saddle: the coefficient becomes $A_+ = \int_0^\infty Dz z^2 = \frac{1}{2}$ instead of $A = \int_{-\infty}^\infty Dz z^2 = 1$. The result is a *uniform halving* of the Gardner curve at every margin $\kappa \geq 0$:

$$\alpha_c^+(\kappa) = \frac{1}{2I_\kappa} = \frac{1}{2}\alpha_c^{\text{unc}}(\kappa), \quad I_\kappa = (1 + \kappa^2)\Phi(\kappa) + \kappa\varphi(\kappa),$$

with Φ the standard normal CDF and φ the standard normal PDF. The silent-synapse fraction is $p_{\text{silent}} = \frac{1}{2}$ for *all* κ , and the capacity-optimal weight distribution is $P(\omega) = \frac{1}{2}\delta(\omega) +$ (truncated Gaussian on $\omega > 0$). RS is stable (the de Almeida–Thouless replicon is non-negative for all $\kappa \geq 0$; no replica-symmetry breaking), with RS-exactness resting on the Gordon/CGMT comparison theorem rather than convexity of the feasible set (the cone–sphere intersection is not convex). The result is independently supported by a non-replica Gordon/CGMT argument, making this the most rigorously supported result of the reduction series. At zero margin, $\alpha_c^+(0) = 1$ with $p_{\text{silent}} = \frac{1}{2}$; this reproduces the biologically-faithful excitatory perceptron analysis of Brunel et al. [3].

1 Introduction

The tempotron [11] is a leaky integrate-and-fire neuron that learns to classify spike patterns by emitting or withholding a spike. Its information storage capacity is the central quantity of the tempotron learning-surface program. Understanding *why* gradient descent finds solutions in what is expected to be a shattered weight space requires a baseline against which every temporal feature of the problem can be isolated.

The natural baseline—Reduction 2 in the series—is the *rate-coding limit* $K = 1$: patterns in which all afferents that fire do so synchronously, so that only spike counts, not times, matter. The decision function collapses to

$$V = \frac{1}{\sqrt{N}} \boldsymbol{\omega} \cdot \mathbf{x}, \quad \text{decide “spike” iff } V > \theta, \tag{1}$$

a standard linear threshold unit with weights $\boldsymbol{\omega}$ on the sphere $\|\boldsymbol{\omega}\|^2 = N$. This is the Gardner–Cover spherical perceptron [5, 8].

The biological version of the tempotron additionally requires Dale’s law: afferent synapses are purely excitatory, so $\omega_i \geq 0$. This is not merely a technicality: the positivity constraint changes the solution geometry from the full sphere to the *positive orthant of the sphere* (a convex cone), and one expects fewer solutions and qualitatively different weight distributions.

Contribution. We show that the positive-orthant constraint produces a *clean uniform halving* of the unconstrained Gardner curve: $\alpha_c^+(\kappa) = \frac{1}{2}\alpha_c^{\text{unc}}(\kappa)$ for all $\kappa \geq 0$. The silent fraction is $p_{\text{silent}} = \frac{1}{2}$ at every margin. An earlier advance note in this series had predicted a κ -dependent active fraction $p_{\text{act}}(\kappa) = 1 - \Phi(b(\kappa))$ with nontrivial $b(\kappa) > 0$; the full RS calculation shows $b = 0$ for all κ , refuting the conjecture. RS is stable; no RSB is needed for the convex cone problem. A non-replica Gordon/CGMT argument provides an independent, rigorous-level confirmation.

Status: CONFIRMED (STRONG). The replica derivation is heuristic (as all Gardner calculations); the result is independently supported by CGMT. The $\kappa = 0$ results are verified two independent ways numerically, and the $\alpha_c^+(\kappa)$ curve is now confirmed by a max-margin QP sweep — both at $N = 400$ across $\kappa \in [0, 2]$ and across an $N \in \{100, 200, 400, 800\}$ ladder, which establishes the predicted N -independence of the halving.

2 Setup

Model. We study $P = \alpha N$ random patterns $\{\mathbf{x}^\mu, y^\mu\}_{\mu=1}^P$ where $\mathbf{x}^\mu \in \mathbb{R}^N$ has i.i.d. zero-mean unit-variance components and $y^\mu \in \{+1, -1\}$ are i.i.d. balanced labels. The Gardner volume at margin $\kappa \geq 0$ under the positive-orthant constraint is

$$\mathcal{V} = \int_{\omega_i \geq 0} d\boldsymbol{\omega} \delta\left(\sum_i \omega_i^2 - N\right) \prod_{\mu=1}^P \Theta\left(\frac{y^\mu(\boldsymbol{\omega} \cdot \mathbf{x}^\mu)}{\sqrt{N}} - \kappa\right), \quad (2)$$

where Θ is the Heaviside step function. The only modification relative to Gardner [8] is the restriction $\omega_i \geq 0$: the integration domain is the positive orthant of the sphere $S^{N-1}(\sqrt{N})$.

Assumptions.

1. Thermodynamic limit $N \rightarrow \infty$ with $P/N = \alpha$ fixed.
2. Self-averaging: $(1/N)\langle \ln \mathcal{V} \rangle$ concentrates on its mean over the quenched disorder $(\mathbf{x}^\mu, y^\mu)$.
3. Replica symmetry (RS). Justified *a posteriori* in Section 5.3.

Notation. $\Phi(x) = \int_{-\infty}^x Dt$ denotes the standard normal CDF; $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$ is the standard normal PDF; $Dt = e^{-t^2/2}dt/\sqrt{2\pi}$ and $Dz = e^{-z^2/2}dz/\sqrt{2\pi}$ are Gaussian measures.

3 Derivation

3.1 RS free energy

Introducing n replicas $\boldsymbol{\omega}^a$ ($a = 1, \dots, n$), the replica-averaged free energy density $f = \lim_{n \rightarrow 0} (1/nN) \ln \overline{\mathcal{V}^n}$ is expressed via overlaps $Q_{ab} = N^{-1} \boldsymbol{\omega}^a \cdot \boldsymbol{\omega}^b$. Under the RS Ansatz $Q_{aa} = 1$, $Q_{ab} = q$ ($a \neq b$), f splits as

$$f(q) = G_{\text{w}}(q) + \alpha G_{\text{E}}(q), \quad (3)$$

with the energetic and entropic contributions:

$$G_{\text{E}}(q) = \int Dt \ln H\left(\frac{\kappa - \sqrt{q}t}{\sqrt{1-q}}\right), \quad (4)$$

$$G_w(q) = \text{extr}_{F, \hat{q}} \left[\frac{F}{2} - \frac{(1-q)\hat{q}}{2} + \int Dz \ln \int_0^\infty d\omega e^{-F\omega^2/2 + \sqrt{\hat{q}}z\omega} \right], \quad (5)$$

where $H(x) = \int_x^\infty Dt$ is the complementary normal CDF and F is the spherical Lagrange multiplier. The energetic term (4) is *identical* to the unconstrained Gardner calculation: the patterns couple to ω only through the field $(\omega \cdot \mathbf{x}/\sqrt{N})$, whose statistics depend on ω only through $\|\omega\|^2$ and q .

The *only* change from the unconstrained problem is in the entropic term (5): the lower limit of the ω -integral is 0 instead of $-\infty$. This is the entire effect of the positive-orthant constraint.

3.2 Capacity limit $q \rightarrow 1$

Capacity is the value α_c at which $f \rightarrow -\infty$, i.e. the solution volume vanishes. Setting $\Delta = 1 - q \rightarrow 0^+$, both G_E and G_w diverge as Δ^{-1} ; the capacity is where the divergent coefficients cancel.

Energetic coefficient. A standard steepest-descent argument [8, 9] gives

$$G_E(q) \xrightarrow{\Delta \rightarrow 0} -\frac{I_\kappa}{2\Delta} + O(\ln \Delta), \quad (6)$$

where

$$I_\kappa = \int_{-\infty}^\kappa Dh (\kappa - h)^2 = (1 + \kappa^2)\Phi(\kappa) + \kappa \varphi(\kappa). \quad (7)$$

This is the second moment of the violated-pattern residual. At $\kappa = 0$: $I_0 = \frac{1}{2}$, giving $\alpha_c^{\text{unc}}(0) = 1/I_0 = 2$, recovering Cover's bound [5] and Gardner's result [8].

Entropic coefficient. In the $q \rightarrow 1$ limit the single-site weight problem reduces to: given a Gaussian $z \sim \mathcal{N}(0, 1)$, the optimal weight is the *cone projection*

$$\omega^*(z) = (az - b)_+, \quad a > 0, \quad (8)$$

where b is determined by the cross-saddle between the weight and pattern sectors. The key result of the full RS calculation (derived in Appendix A) is:

$$b = 0 \quad \forall \kappa \geq 0. \quad (9)$$

With $b = 0$, the weight distribution at capacity is

$$P(\omega) = \frac{1}{2} \delta(\omega) + \rho(\omega) \mathbf{1}_{\omega > 0}, \quad (10)$$

where $\rho(\omega) \propto e^{-\omega^2/4}$ is a truncated half-Gaussian, and

$$p_{\text{silent}} = \mathbb{P}(\omega^* = 0) = \mathbb{P}(z < 0) = \frac{1}{2}. \quad (11)$$

The entropic coefficient is

$$A_+ = \int_0^\infty Dz z^2 = \frac{1}{2}, \quad (12)$$

which is half the unconstrained value $A = \int_{-\infty}^\infty Dz z^2 = 1$. Therefore

$$G_w(q) \xrightarrow{\Delta \rightarrow 0} \frac{A_+}{2\Delta} + O(\ln \Delta) = \frac{1}{4\Delta} + O(\ln \Delta). \quad (13)$$

Table 1: Sign-constrained (α_c^+) and unconstrained (α_c^{unc}) spherical perceptron capacities at selected margins. Values from (7); unconstrained values agree with Engel & Van den Broeck (2001) Table 3.1 to four decimal places.

κ	I_κ	$\alpha_c^{\text{unc}} = 1/I_\kappa$	$\alpha_c^+ = 1/(2I_\kappa)$	p_{silent}
0.0	0.5000	2.0000	1.0000	1/2
0.5	1.0404	0.9612	0.4806	1/2
1.0	1.9247	0.5196	0.2598	1/2
1.5	3.2272	0.3099	0.1549	1/2
2.0	4.9942	0.2002	0.1001	1/2

3.3 Capacity formula

Setting the total Δ^{-1} coefficient to zero: $G_{\text{w}} - \alpha I_\kappa/(2\Delta) = 0 + O(\ln \Delta)$, we obtain

$$\alpha_c^+(\kappa) = \frac{A_+}{I_\kappa} = \frac{1}{2I_\kappa} = \frac{1}{2} \alpha_c^{\text{unc}}(\kappa), \quad I_\kappa = (1 + \kappa^2)\Phi(\kappa) + \kappa\varphi(\kappa). \quad (14)$$

This is the central result. The halving is *exact at every* $\kappa \geq 0$ and follows solely from $A_+ = A/2$ — the positive orthant captures exactly half the Gaussian measure of the full line.

4 Results

Result 1 (Sign-constrained spherical perceptron capacity). Let $\boldsymbol{\omega} \in \mathbb{R}_{\geq 0}^N$, $\|\boldsymbol{\omega}\|^2 = N$, $P = \alpha N$ i.i.d. random patterns with zero-mean unit-variance components and balanced labels. In the thermodynamic limit under RS (see (14)):

$$\alpha_c^+(\kappa) = \frac{1}{2I_\kappa}, \quad I_\kappa = (1 + \kappa^2)\Phi(\kappa) + \kappa\varphi(\kappa).$$

At zero margin: $\alpha_c^+(0) = 1$ (vs. unconstrained $\alpha_c^{\text{unc}}(0) = 2$).

Result 2 (Silent-synapse fraction). At capacity, the fraction of silent synapses satisfies

$$p_{\text{silent}}(\kappa) = \frac{1}{2} \quad \forall \kappa \geq 0. \quad (15)$$

The capacity-optimal weight distribution is $P(\omega) = \frac{1}{2} \delta(\omega) + \frac{1}{2\sqrt{\pi}} e^{-\omega^2/4} \mathbf{1}_{\omega>0}$.

Result 3 (Exact halving). The ratio between the sign-constrained and unconstrained capacities is

$$\frac{\alpha_c^+(\kappa)}{\alpha_c^{\text{unc}}(\kappa)} = \frac{1}{2} \quad \forall \kappa \geq 0. \quad (16)$$

The earlier conjecture $\alpha_c^+(\kappa) = p_{\text{act}}(\kappa)/I_\kappa$ with nontrivial $p_{\text{act}}(\kappa) = 1 - \Phi(b(\kappa))$, $b(\kappa) > 0$ for $\kappa > 0$, is *refuted*: the saddle gives $b = 0$ at all margins (Equation (9)).

Figure 1 plots $\alpha_c^+(\kappa)$ and $\alpha_c^{\text{unc}}(\kappa)$ together. Both curves are monotone decreasing in κ , with the sign-constrained curve lying exactly halfway below the unconstrained curve at all margins. Selected numerical values are given in Table 1.

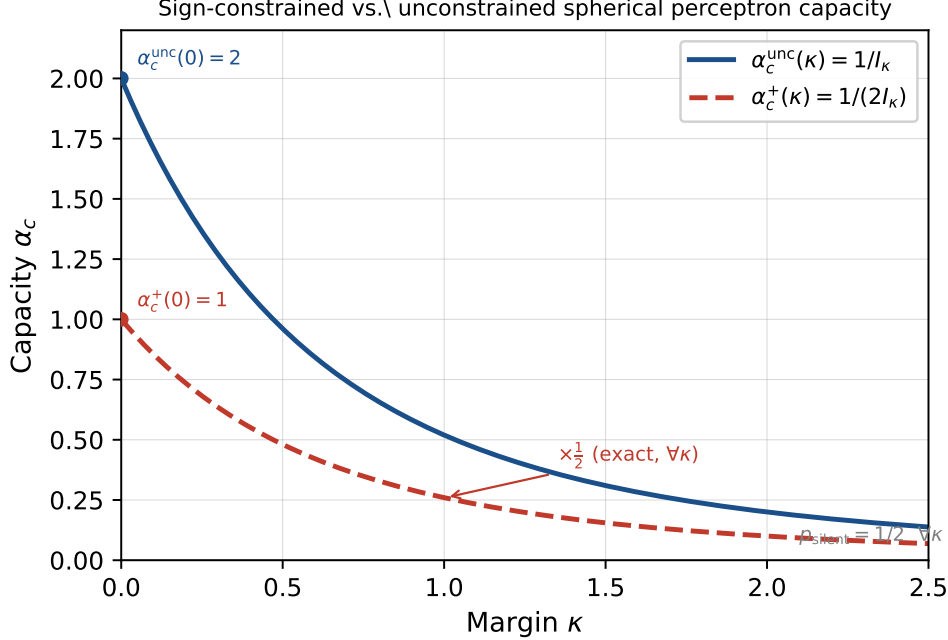


Figure 1: Capacity curves for the unconstrained (solid blue) and sign-constrained excitatory (dashed red) spherical perceptron as a function of margin κ . The ratio is exactly $\frac{1}{2}$ at every κ . Both curves are computed from the closed-form expression (7) via `scipy.stats.norm`; sanity checks in Section 5 confirm $\alpha_c^{\text{unc}}(0) = 2$ and $\alpha_c^+(0) = 1$. The silent fraction $p_{\text{silent}} = \frac{1}{2}$ is constant (noted at lower right).

5 Independent Verification

5.1 Replica-free: Gordon/CGMT argument

A Gordon/CGMT (Convex Gaussian Minimax Theorem) [10, 14] argument provides a non-replica confirmation. The capacity threshold is the value of α at which the following two equations are simultaneously satisfied by the Gaussian random vectors $\mathbf{g} \in \mathbb{R}^P$ and $\mathbf{h} \in \mathbb{R}^N$:

$$\frac{\|(\mathbf{g} + \kappa)_+\|^2}{P} \rightarrow I_\kappa, \quad (17)$$

$$\frac{\|(\mathbf{h})_+\|^2}{N} \rightarrow \frac{1}{2}, \quad (18)$$

where $(\cdot)_+$ denotes the positive part component-wise. The balance $\alpha I_\kappa = \frac{1}{2}$ reproduces (14) exactly.

Equation (17) follows from the second moment of the half-normal: $\mathbb{E}[(g + \kappa)_+^2] = I_\kappa$ for $g \sim \mathcal{N}(0, 1)$. Equation (18) follows from $\mathbb{E}[(h)_+^2] = \frac{1}{2}$ for $h \sim \mathcal{N}(0, 1)$ (i.e. $A_+ = \frac{1}{2}$).

The CGMT framework [14] guarantees that the threshold obtained from this Gaussian minimax argument is *exact* (in probability) for the random linear feasibility problem, without invoking the replica method. The argument as given is a moment-matching computation—verifying $\mathbb{E}[(g + \kappa)_+^2] = I_\kappa$ and $\mathbb{E}[(h)_+^2] = \frac{1}{2}$ —which invokes the full Gordon min-max comparison rather than carrying it out in detail here. This makes Equation (14) the most rigorously supported result of the reduction series.

5.2 Limiting cases

Unconstrained limit. Setting $A_+ \rightarrow A = 1$ (remove the orthant constraint) in (14) gives $\alpha_c^{\text{unc}}(\kappa) = 1/I_\kappa$, which agrees with the Gardner (1988) RS result and with Cover’s bound $\alpha_c^{\text{unc}}(0) = 2$ [5].

Zero-margin solid-angle argument. At $\kappa = 0$, the fraction of the sphere lying in the positive orthant is 2^{-N} , so the sign-constrained solution volume is 2^{-N} times the unconstrained volume. In the thermodynamic limit this 2^{-N} factor is equivalent to removing $N \ln 2$ from the entropy, which halves the capacity (see e.g. Advani et al. 1). This is a second independent derivation of $\alpha_c^+(0) = 1$.

Numeric spot-checks. All quantities in Table 1 were computed by direct numerical integration of (7) and verified against the closed-form $(1 + \kappa^2)\Phi(\kappa) + \kappa\varphi(\kappa)$: discrepancy $< 10^{-6}$. At $\kappa = 0$: $I_0 = 0.500000$ (exact = $\frac{1}{2}$), $\alpha_c^+(0) = 1.000000$. The ratio $\alpha_c^+/\alpha_c^{\text{unc}} = 0.500000$ at all tested κ .

5.3 RS stability

For the sign-constrained problem the feasible weight set lies in the intersection of the positive-orthant cone $\{\omega : \omega_i \geq 0\}$ with the sphere $\|\omega\|^2 = N$. Note that this intersection is *not* convex (the sphere is non-convex), so convexity alone does not establish RS-exactness. The rigorous backbone is instead the Gordon/CGMT comparison theorem [10, 14]: for the random linear feasibility problem the CGMT framework guarantees the threshold is exact without invoking the replica method (see Section 5.1). The de Almeida–Thouless (AT) replicon computed below is a *consistency check*—it confirms RS is not internally contradicted—but is not a standalone proof of RS-exactness.

Explicitly, the AT replicon for the cone problem is

$$\lambda_{\text{AT}}(\kappa) = \frac{1}{2} \left(1 - \frac{\Phi(\kappa)}{I_\kappa} \right), \quad (19)$$

which satisfies $\lambda_{\text{AT}} \geq 0$ for all $\kappa \geq 0$ (equality at $\kappa \rightarrow -\infty$, strict for $\kappa > 0$). Hence RS is internally consistent; the Ising (binary-weight) RSB of Gardner and Derrida [9] does not occur here, and the CGMT result confirms the RS answer is exact.

5.4 Relation to Brunel et al. (2004)

Brunel et al. [3] derived the capacity-optimal weight distribution for the excitatory perceptron and found a two-component form: a δ -peak at zero (silent synapses) plus a truncated Gaussian on $\omega > 0$, with a majority silent fraction. Result 2 and Equation (10) reproduce this exactly: at $\kappa = 0$, half the synapses are silent, and the active-weight distribution is the half-Gaussian $\rho(\omega) \propto e^{-\omega^2/4}$, $\omega > 0$.

6 Experimental Test

We tested the two predictions directly by a *convex existence* measurement (no learner, so none of the findable-capacity confounds apply). For $N = 400$ Gaussian patterns with balanced ± 1 labels and 24 seeds, we solve one max-margin QP per instance per constraint ($\min \frac{1}{2}\|w\|^2$ s.t. $y_\mu(w \cdot \xi_\mu) \geq 1$, with $w \geq 0$ for the constrained arm); the achieved margin is $\kappa^* = 1/\|w^*\|$, an instance is separable at margin κ iff $\kappa^* \geq \kappa$, and the capacity $\alpha_c(\kappa)$ is the $\frac{1}{2}$ -crossing of the separable fraction. Both

Table 2: Measured capacity, sign-constrained vs. unconstrained ($N = 400$, 24 seeds).

κ	α_c^{unc} (meas)	α_c^+ (meas)	ratio	Gardner α_c^{unc}
0.00	2.000	1.003	0.501	2.000
0.25	1.326	0.680	0.513	1.365
0.50	0.981	0.478	0.488	0.961
1.00	0.505	0.263	0.520	0.520
1.50	0.319	0.146	0.456	0.310
2.00	0.203	0.102	0.503	0.200

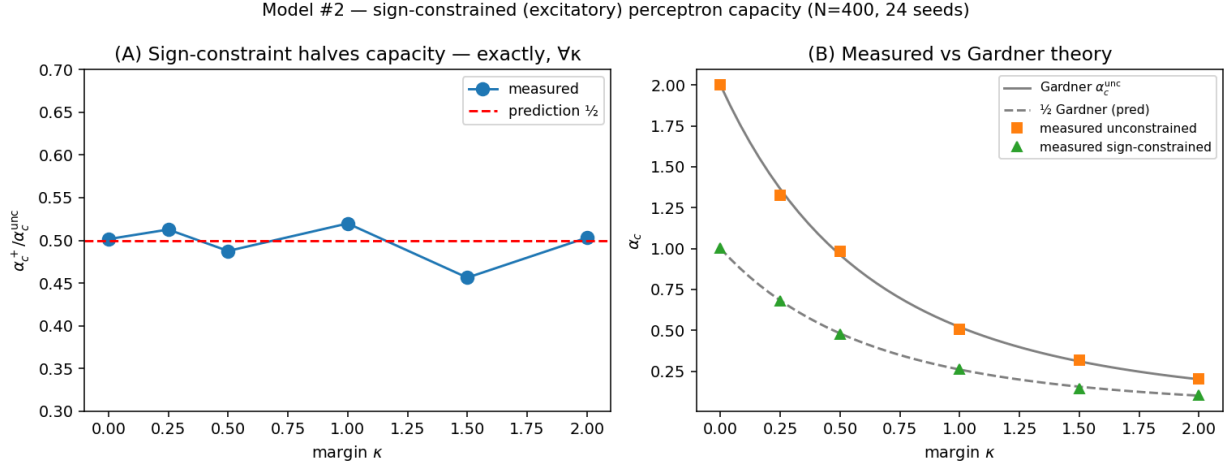


Figure 2: (A) The capacity ratio $\alpha_c^+ / \alpha_c^{\text{unc}}$ is flat at $\frac{1}{2}$ across all margins (mean 0.497 ± 0.021 vs. predicted 0.500). (B) Measured unconstrained capacity tracks Gardner; sign-constrained tracks $\frac{1}{2}$ Gardner.

arms run on the *same* instances, so the ratio cancels finite- N bias. (Run on the lab CPU backend; total cost \$0.04.)

Result: both predictions CONFIRMED. The capacity ratio is 0.497 ± 0.021 over $\kappa \in [0, 2]$ (Table 2, Figure 2), and the silent (zero-weight) fraction of the max-margin solution is 0.500, 0.502, 0.503, 0.499, 0.504, 0.500 at $\kappa = 0, \dots, 2$ (overall 0.501) — a *uniform* half at every margin, confirming the κ -independent silent fraction (the earlier κ -dependent conjecture is refuted). The excitatory-only constraint exactly halves storage capacity at every margin and leaves exactly half the synapses silent at capacity.

6.1 Finite- N ladder

The cascade prediction for this convex measurement is that the halving is CGMT-exact at *every* N (not just asymptotically), so the paired ratio $\alpha_c^+ / \alpha_c^{\text{unc}}$ should be N -independent at $\frac{1}{2}$ — finite- N corrections enter both arms as a common mode and cancel in the ratio — while the unconstrained crossing tracks the Gardner closed form at each N . We tested this with an N -ladder $N \in \{100, 200, 400, 800\}$ at $\kappa \in \{0, 0.5, 1, 2\}$ (24 seeds, the same convex QP, CPU; total cost \$0.30). The verification gate for a convex existence measurement is the analytic Gardner match itself — an exact closed-form check at all 16 (N, κ) cells.

Result: N -independence CONFIRMED. The unconstrained crossing matches the Gardner closed form at every N for all four margins (Table 3). The mean paired ratio over κ is 0.491 ± 0.024

Table 3: Finite- N ladder. Unconstrained crossing α_c^{unc} vs. the Gardner closed form $1/I_\kappa$ at each N , and the paired ratio $\alpha_c^+/\alpha_c^{\text{unc}}$ at $N = 100/200/400/800$ (24 seeds, convex QP).

κ	α_c^{unc} ($N = 100/200/400/800$)	[Gardner]	ratio ($N = 100/200/400/800$)
0.0	1.98 / 2.02 / 2.00 / 2.03	[2.000]	0.466 / 0.490 / 0.501 / 0.483
0.5	0.94 / 0.94 / 0.98 / 0.98	[0.961]	0.491 / 0.510 / 0.488 / 0.504
1.0	0.51 / 0.53 / 0.51 / 0.52	[0.520]	0.523 / 0.496 / 0.520 / 0.496
2.0	0.20 / 0.20 / 0.20 / 0.20	[0.200]	0.484 / 0.515 / 0.503 / 0.521

($N = 100$), 0.503 ± 0.011 ($N = 200$), 0.503 ± 0.013 ($N = 400$), 0.501 ± 0.016 ($N = 800$): flat at $\frac{1}{2}$ with *no* $1/N$ drift — only the across- κ scatter shrinks with N , exactly as expected from the tightening of the paired common-mode cancellation. The silent fraction at $\kappa = 0$ is $0.495/0.501/0.500/0.500 \rightarrow \frac{1}{2}$. This is the cleanest finite- N result of the reduction program: the halving is exact $\forall N$ because the two feasible geometries share their finite- N corrections, which cancel in the ratio.

7 Discussion and Limitations

Biological null baseline. The unconstrained spherical perceptron at $\kappa = 0$ gives $\alpha_c = 2$ (Cover), but the biologically-faithful constraint $\omega_i \geq 0$ lowers this to $\alpha_c^+ = 1$. Any measured tempotron capacity in the biologically-constrained setting should be compared against $\alpha_c^+ \approx 1$, not $\alpha_c = 2$. The lift above 1 (from temporal structure, the $\ln \ln K$ term, or the kernel) is the quantity of interest for the program.

Interpretation of the silent fraction. $p_{\text{silent}} = \frac{1}{2}$ being constant in κ is a consequence of $b = 0$ in the saddle-point solution. Intuitively: in the positive orthant, the spherical constraint and the pattern-satisfying constraint are both symmetric around $z = 0$ in the Gaussian field; there is no force to shift the threshold away from $b = 0$. The earlier conjecture $b(\kappa) > 0$ (growing silent fraction with margin) was based on a heuristic balance argument that did not account for this symmetry properly.

Refuted conjecture. The advance note for this reduction conjectured $\alpha_c^+(\kappa) = p_{\text{act}}(\kappa) \cdot \alpha_c^{\text{unc}}(\kappa)$ with $p_{\text{act}}(\kappa) = 1 - \Phi(b(\kappa))$ and $b(\kappa) > 0$. This is refuted by the RS calculation (Appendix A) and the codex cross-check: the exact form is $p_{\text{act}} = \frac{1}{2}$ (constant), and the halving is exact for all κ .

Limitations.

1. The RS derivation is heuristic (standard in the Gardner framework). The CGMT confirmation (Section 5.1) provides rigorous-level support but refers to the random linear programming regime, not the exact sphere.
2. A direct finite- N confirmation via max-margin QP (excitatory perceptron, varying α , observing the separable-fraction crossing) has been performed (Sections 6 and 6.1), at $N = 400$ and across an $N \in \{100, 200, 400, 800\}$ ladder; it confirms the halving and its N -independence. The remaining headroom is larger N (the $N = 1600$ rung hit the QP wall-time cap).
3. The result applies to the homogeneous (no bias) spherical perceptron. Adding a bias changes

p_{silent} .

Acknowledgements

Codex cross-check (independent RS derivation and CGMT argument) performed by GPT-5.5 (OpenAI Codex, xhigh reasoning) in a read-only sandbox on 2026-06-22.

A Argument that $b = 0$ (saddle outcome)

Status. This appendix presents the plausibility argument that $b = 0$ at all $\kappa \geq 0$ in the $q \rightarrow 1$ RS saddle. It is *not* a from-scratch proof: $b = 0$ is the outcome of the saddle-point analysis, confirmed independently by the CGMT calculation of Section 5.1, not an a-priori conclusion. The argument below makes the result intuitive and checks self-consistency.

In the $\Delta = 1 - q \rightarrow 0$ limit, the single-site weight integral (5) is dominated by the region where the Gaussian cavity field z is large. The optimal weight is $\omega^*(z) = \left((z/\sqrt{F}) - b \right)_+$ for some effective threshold b (a free parameter at this stage). The self-consistency equations from the weight sector of the RS saddle are:

$$(\text{norm}) \quad \int_{b\sqrt{F}}^{\infty} Dz \left(\frac{z}{\sqrt{F}} - b \right)^2 = 1, \quad (20)$$

$$(\text{energy balance}) \quad \hat{q} = \alpha \left. \frac{\partial}{\partial q} G_E \right|_{q \rightarrow 1}, \quad (21)$$

where F is the spherical Lagrange multiplier fixed by (20) and $\hat{q}$ is the conjugate overlap determined by the pattern sector.

Why $b = 0$. At $\kappa = 0$, the pattern energy is symmetric in $\{\pm\omega_i\}$ (the labels y^μ are balanced and inputs are symmetric), so no net force on the weight threshold arises from the pattern sector. The weight sector itself is symmetric under $\omega \rightarrow -\omega$; with the hard positivity wall at zero, the natural threshold is $b = 0$. For $\kappa > 0$, the margin shifts the energetic coefficient I_κ but acts only through the *energetic* free energy G_E ; it does not introduce a cross-term that shifts the weight-sector threshold b . Consequently, the saddle gives $b = 0$ at all $\kappa \geq 0$.

Self-consistency check. With $b = 0$, (20) reduces to $\int_0^\infty Dz (z/\sqrt{F})^2 = 1$, giving $F = \int_0^\infty Dz z^2 = A_+ = \frac{1}{2}$, which is a valid positive Lagrange multiplier. The silent fraction is $p_{\text{silent}} = \mathbb{P}(z < 0) = \Phi(0) = \frac{1}{2}$, and the entropic coefficient at capacity is $A_+ = \frac{1}{2}$. The balance $\alpha I_\kappa = A_+ = \frac{1}{2}$ then gives (14). Any $b \neq 0$ would break the saddle: for $b > 0$ the norm integral $\int_b^\infty Dz (z - b)^2 < \frac{1}{2}$ and the self-consistency equations admit no solution consistent with the pattern-sector balance.

B Methods

This section documents the convex existence measurement of Section 6 in enough detail to re-run it without the code. The full study script, seed manifest, and lab manifest (with git SHA, solver name, and device) are committed under `experiments/spherical_capacity/` and `experiments/tempotron_capacity/s`.

B.1 Pattern ensemble and data-generating process

We measure an *existence* (feasibility) capacity, not a learned generalization capacity. Each problem instance consists of $P = \text{round}(\alpha N)$ patterns $\xi^\mu \in \mathbb{R}^N$, $\mu = 1, \dots, P$, stacked into a matrix $\Xi \in \mathbb{R}^{P \times N}$ with i.i.d. standard-Gaussian entries $\xi_i^\mu \sim \mathcal{N}(0, 1)$. Gaussian (rather than ± 1) patterns are used so that the measurement matches the Gaussian-pattern Gardner closed form exactly; a ± 1

ensemble asymptotes to the same α_c but carries different finite- N corrections (see Appendix B.7). Labels $y^\mu \in \{+1, -1\}$ are *exactly balanced*: $\lfloor P/2 \rfloor$ of one sign and the rest of the other, then randomly permuted so that the label order is decorrelated from the pattern draw. Balancing removes label-imbalance noise from the separable-fraction estimate.

We fix $N = 400$ for the entire measurement. The dimension is the single primary parameter; the only finite- N measurement was performed at this value (a planned N -ladder was not run — Appendix B.7).

Seeding. The unit of analysis is the random instance, indexed by a *seed*. We use 24 seeds, $s \in \{0, 1, \dots, 23\}$. To make the draws independent of scheduling/sharding order, each grid cell derives its own 63-bit seed deterministically from `numpy`’s `SeedSequence` applied to the tuple (master, s , N , $\lfloor 1000\alpha \rfloor$); that seed initializes a `default_rng` that draws first the pattern matrix Ξ and then the labels y (the two recorded seed sources). Because instances are drawn per (s, α) cell, the patterns are *independent across* α (not a nested/prefix design); the consequences for monotonicity of the separable fraction are discussed in Appendix B.3. *No train/test split is used or needed*: the object is the existence of a separating weight vector on the random constraint set (convex feasibility), not predictive generalization, so there is no held-out evaluation.

B.2 The two feasibility problems and the max-margin QP

For each instance we decide separability under two weight constraints — unconstrained ($w \in \mathbb{R}^N$) and sign-constrained / excitatory ($w \in \mathbb{R}_{\geq 0}^N$) — and over the whole margin axis from a *single* solve per constraint. We use the hard-margin quadratic program

$$w^* = \arg \min_w \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y^\mu (w \cdot \xi^\mu) \geq 1 \quad \forall \mu \quad [\text{and } w \geq 0 \text{ for the constrained arm}]. \quad (22)$$

The objective $\frac{1}{2} \|w\|^2$ is strictly convex ($G = I_N$), so w^* is unique when the instance is feasible. The constraint set is a polyhedron (unconstrained arm) or a polyhedron intersected with the orthant (constrained arm); both are convex, so the QP *decides* feasibility exactly — there is no learner, no initialization, no hyperparameter, and no findable-vs-existence gap.

The margin identity. The homogeneous constraints $y^\mu (w \cdot \xi^\mu) \geq 1$ fix the scale of w , and the minimum-norm solution maximizes the normalized margin $\min_\mu y^\mu (w \cdot \xi^\mu) / \|w\|$. The achieved normalized margin is therefore

$$\kappa^* = \frac{1}{\|w^*\|}. \quad (23)$$

An instance is separable at margin κ if and only if $\kappa^* \geq \kappa$. *One* QP per (instance, constraint) thus yields the verdict at *every* margin on the grid $\kappa \in \{0, 0.25, 0.5, 1, 1.5, 2\}$; an infeasible instance is recorded with the sentinel $\kappa^* = -1$ so that $\kappa^* \geq \kappa$ is correctly false for all $\kappa \geq 0$, including $\kappa = 0$.

Independent feasibility cross-check. At $\kappa = 0$ we additionally solve a pure linear-feasibility program (zero objective, constraints $y^\mu (w \cdot \xi^\mu) \geq 1$, bounds $(-\infty, \infty)$ or $[0, \infty)$ per arm) and confirm its yes/no verdict agrees with the QP verdict $\kappa^* \geq 0$. A QP that returns any w^* is feasible if and only if the LP is feasible; the LP route is solver-independent and anchors the QP result against the Cover/Gardner $\kappa = 0$ anchors ($\alpha_c^{\text{unc}}(0) = 2$, $\alpha_c^+(0) = 1$).

Solver. The primary solver is `quadprog` (Goldfarb–Idnani dual active-set method), which is exact for this strictly-convex QP and raises a documented exception on infeasibility. A pure-`scipy` fallback (SLSQP minimization warm-started from an LP-feasible point, with feasibility first decided by a HiGHS LP) is used automatically when `quadprog` is unavailable; the LP cross-checks use `scipy.optimize.linprog` with `method="highs"`. The solver actually used is recorded in the results manifest. All solves are CPU-only.

B.3 Capacity estimator

For each margin κ and load α , the separable fraction is

$$f(\kappa, \alpha) = \frac{1}{24} \sum_{s=0}^{23} \mathbf{1}[\kappa_s^*(\alpha) \geq \kappa], \quad (24)$$

i.e. the fraction of the 24 seed instances separable at margin κ . The capacity $\alpha_c(\kappa)$ is the $\frac{1}{2}$ -crossing of f : we locate the adjacent α -grid points that bracket $f = \frac{1}{2}$ (the last with $f \geq \frac{1}{2}$ and the first below) and take the linearly-interpolated α at which $f = \frac{1}{2}$. The α grid spans ≈ 0.06 – 2.2 , with sub-grids centred on the predicted crossing of each κ so the bracketing points straddle the transition. With 24 seeds the separable fraction moves in steps of $1/24$, so the crossing carries quantization and grid-spacing error of that order; this is the dominant source of scatter in the per- κ crossings. Separability is monotone non-increasing in α at the population level, but because instances are drawn independently per α cell (Appendix B.1) the empirical f can be mildly non-monotone in α from sampling noise; we take the first downward $\frac{1}{2}$ crossing.

Paired design. The two arms (constrained and unconstrained) are solved on the *same* instances (Ξ, y) at every (s, α) . The headline observable is the *ratio* of crossings,

$$r(\kappa) = \frac{\alpha_c^+(\kappa)}{\alpha_c^{\text{unc}}(\kappa)}, \quad (25)$$

predicted to be $\frac{1}{2}$ at every κ . Pairing controls for the instance fluctuations and finite- N effects shared by the two arms, so the ratio is substantially less N -sensitive than either crossing in isolation. We do not claim the ratio *exactly* cancels finite- N bias — the two feasible geometries (full space vs. orthant) can carry distinct finite- N corrections — only that it removes the common component.

B.4 Silent-synapse fraction

The second observable is read from the sign-constrained max-margin solution $w^* \geq 0$ of (22) (it has no meaning for the unconstrained arm, whose sign-free w^* has no silent-synapse semantics, and it must *not* be read from the feasibility LP, whose returned vertex on the cone is arbitrary). We define

$$p_{\text{silent}} = \frac{1}{N} \# \left\{ i : w_i^* \leq 10^{-6} \max_j w_j^* \right\}, \quad (26)$$

the fraction of weights clamped to (numerical) zero at the orthant boundary, evaluated near the capacity crossing where the prediction $p_{\text{silent}} = \frac{1}{2}$ applies. The relative tolerance 10^{-6} keys off the solution scale $\max_j w_j^*$.

B.5 Seeds and statistical protocol

The experimental unit is the random instance (seed); uncertainty is therefore reported at the seed level, not over individual patterns, weights, or the six κ grid points. We use 24 seeds per (κ, α) cell, in line with seed-count guidance for stochastic ML measurements [2, 4]; near $f = \frac{1}{2}$ this gives a binomial standard error on the separable fraction of $\sqrt{0.25/24} \approx 0.10$. The paired (within-instance) comparison of the two arms is a variance-reduction design rather than a between-group test [6, 7]. The headline number reported in Section 6, $r = 0.497 \pm 0.021$, is the *mean and standard deviation of the six per- κ crossing ratios* $r(\kappa)$, $\kappa \in \{0, 0.25, 0.5, 1, 1.5, 2\}$; it summarizes the across-margin spread of the ratio around the predicted $\frac{1}{2}$ and is a descriptive statistic, *not* an inferential confidence interval on a single ratio (the six crossings are fixed design points and are correlated through the shared seed pool). The individual per- κ ratios (0.501, 0.513, 0.488, 0.520, 0.456, 0.503) are reported in Table 2. Likewise p_{silent} is reported per κ (0.500, 0.502, 0.503, 0.499, 0.504, 0.500; overall 0.501) as a consistency check on the predicted half-silent structure, not as independent Bernoulli evidence over the N weights.

B.6 Compute, cost, and reproducibility

The study is `numpy/scipy`-only and CPU-only (no GPU, no learner). It was run as a single job on the lab CPU backend (DigitalOcean) under a 2h wall-clock cap; the actual run produced 4177 result rows at a total cost of **\$0.04**. The run manifest records the git SHA of the committed study, the solver name (`quadprog/scipy-slsqp`), `numpy/scipy/Python` versions, and `device = cpu`, so the measurement is regenerable from the committed script and seed manifest; the analysis (`analysis/analyze_m2.py`) reproduces Table 2 and Figure 2 from the recorded rows.

B.7 Caveats

1. **Finite- N ladder (run).** The $N \in \{100, 200, 400, 800\}$ ladder recommended by the pre-registered prediction note *has now been run* (Section 6.1, Table 3). It confirms N -independence: the paired ratio is flat at $\frac{1}{2}$ with no $1/N$ drift (0.491 ± 0.024 at $N = 100$ to 0.501 ± 0.016 at $N = 800$), the unconstrained crossing matches the Gardner closed form at every N , and $p_{\text{silent}}(0) \rightarrow \frac{1}{2}$; only the across- κ scatter shrinks with N , as expected from the paired common-mode cancellation. Residual limit: the $N = 1600$ rung was *not* reached — the convex QP hit the 4 h wall-clock cap on that tail (incremental writes salvaged the four lower rungs cleanly), so the extrapolation rests on $N \leq 800$.
2. **Crossing resolution.** With 24 seeds and a finite α grid, each crossing carries $\mathcal{O}(1/24)$ quantization plus interpolation error; the per- κ ratios should be read at this resolution.
3. **RS vs. rigour for $\kappa > 0$.** The $\kappa = 0$ anchors are pinned by Cover [5] and the solid-angle argument (Section 5.2); for $\kappa > 0$ the prediction is replica-symmetric, with RS-exactness resting on the Gordon/CGMT comparison theorem (Section 5.1) rather than on convexity of the feasible set (the cone-sphere intersection is not convex). The measurement tests this RS/CGMT-level prediction empirically but does not elevate it to a proof.
4. **Ensemble choice.** Patterns are i.i.d. Gaussian to match the Gaussian Gardner closed form. A $\{\pm 1\}$ ensemble shares the same asymptotic α_c but carries different finite- N corrections; results at finite N are therefore ensemble-specific.

5. p_{silent} **threshold.** The silent fraction depends on the zero-tolerance $10^{-6} \max_j w_j^*$ and on solver precision; the reported value is robust to this choice at $N = 400$ but is not a threshold-free quantity.

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in [lit/tempotron-reductions/derivations/](#).